SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – **Analytical problem solving**
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
Student Name:	P. GAYATHRI	**Reg. No.:**	192425277
Section / Batch:	BTECH-AI ML	**Date:**	17-07-2026

Code of Conduct
I _P.GAYATHRI-192425277 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: ____________________

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problem Solving **Duration: 1hr**

Q1.

Design a machine learning solution for predicting customer loan approval using a suitable learning algorithm. **Describe** the major stages involved in model development, including data preparation, model training, and performance evaluation. (10 marks)

ANS:

Introduction

Machine Learning (ML) enables computers to learn from historical data and make predictions without being explicitly programmed. Loan approval prediction is a **supervised classification** problem where the model predicts whether a customer's loan application should be **approved** or **rejected** based on previous loan records.

A suitable algorithm for this problem is the **Random Forest Classifier**, as it provides high accuracy, handles large datasets efficiently, and reduces overfitting.

Stages of Model Development

1. Data Collection

Collect historical customer loan data from banks or financial institutions. Typical features include:

- Age
- Gender
- Income
- Occupation
- Credit Score
- Loan Amount
- Employment Status
- Previous Loan History
- Loan Approval Status (Target Variable)

2. Data Preparation

Data preprocessing improves data quality before training.

Steps include:

- Remove duplicate records.
- Handle missing values.
- Convert categorical variables into numerical form (Encoding).

- Normalize or standardize numerical values.
- Split the dataset into:
    - Training Data (80%)
    - Testing Data (20%)

3. Model Selection
Choose a suitable supervised learning algorithm such as:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Random Forest is preferred because:

- High accuracy
- Handles missing values
- Reduces overfitting
- Works well with large datasets

4. Model Training
The algorithm learns patterns from the training dataset by building multiple decision trees and combining their predictions.

5. Model Prediction
After training, the model predicts whether a new customer's loan will be:

- Approved
- Rejected

based on the input features.

6. Model Evaluation
Evaluate the model using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve

Example:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Advantages

- Faster loan approval process
- Reduces human errors
- Improves prediction accuracy
- Helps banks minimize financial risk

Applications

- Banking
- Credit card approval
- Insurance claim prediction
- Financial risk analysis

Conclusion

A machine learning model using Random Forest can accurately predict loan approvals by learning from historical customer data. Proper data preprocessing, model training, and evaluation improve decision-making and reduce financial losses.

Q2.

Describe the architecture of an Artificial Neural Network (ANN) consisting of input, hidden, and output layers. **Illustrate** the role of artificial neurons, weights, bias, and activation functions in processing input data to generate predictions. (10 marks)

ANS:

Introduction

An Artificial Neural Network (ANN) is a machine learning model inspired by the structure of the human brain. It consists of interconnected artificial neurons that process information and learn patterns from data.

Architecture of ANN

```
Input Layer
   |
   ▼
Hidden Layer
   |
   ▼
```

Output Layer

1. Input Layer

- First layer of the network.
- Receives input features from the dataset.
- Does not perform computation.

Example:

- Age
- Income
- Credit Score

2. Hidden Layer

- Performs mathematical computations.
- Extracts patterns and relationships from the data.
- There may be one or more hidden layers.

Each neuron computes:

$$Z = \sum(W_i X_i) + b$$

Where:

- X = Input
- W = Weight
- b = Bias

3. Output Layer

Produces the final prediction.

Examples:

- Loan Approved / Rejected
- Spam / Not Spam
- Cat / Dog / Bird

Artificial Neuron

Each neuron performs three operations:

1. Receives input
2. Multiplies inputs by weights
3. Applies activation function

Neuron equation:

$$Z = W_1 X_1 + W_2 X_2 + \cdots + W_n X_n + b$$

Output:

$$Y = f(Z)$$

Weights
Weights determine the importance of each input.
Higher weight means greater influence on prediction.

Bias
Bias shifts the activation function and helps improve learning.
Equation:

$$Z = WX + b$$

Activation Functions
Activation functions introduce non-linearity.
Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$

Range: 0 to 1
Used for binary classification.

ReLU

$$f(x) = \max(0, x)$$

Fast and widely used in deep learning.

Tanh
Range: -1 to 1

Softmax
Used for multiclass classification.

Working of ANN
1. Inputs enter the input layer.
2. Hidden layer calculates weighted sum.
3. Activation function generates neuron output.
4. Output layer predicts final result.
5. During training, weights and bias are updated using Backpropagation.

Applications

- Image Recognition
- Speech Recognition
- Medical Diagnosis
- Fraud Detection
- Self-driving Cars

Advantages

- Learns complex patterns
- High prediction accuracy
- Handles nonlinear problems

Conclusion

ANN is one of the most powerful machine learning models. By adjusting weights and biases, it learns complex relationships and provides accurate predictions for classification and regression problems.

Q3.

Analyze the working of the Perceptron learning algorithm for solving a binary classification problem. **Demonstrate** the weight and bias update process for one iteration using a suitable numerical example. (10 marks)

ANS:

Introduction

The Perceptron is the simplest type of Artificial Neural Network used for binary classification. It learns by updating weights whenever a prediction is incorrect.

Perceptron Structure

```
X1 ──W1──\
      \
X2 ──W2────► Neuron ► Output
      /
Bias ─────/
```

Algorithm Steps

Step 1

Initialize weights and bias randomly.

Step 2

Calculate weighted sum.

$$Z = \sum WX + b$$

Step 3
Apply activation function.
If

$$Z \geq 0$$

Output =1
Otherwise
Output =0

Step 4
Calculate error.

$$Error = Target - Predicted$$

Step 5
Update weights.

$$W_{new} = W + \eta(Error)X$$

Bias update:

$$b_{new} = b + \eta(Error)$$

Repeat until all samples are classified correctly.

Numerical Example
Given:
Input:
X=(2,3)
Weights:
W=(0.5,0.2)
Bias=0.1
Learning Rate=0.1
Target=1

Calculate Output

$$Z = (2 \times 0.5) + (3 \times 0.2) + 0.1$$

=1+0.6+0.1
=1.7

Since
Z>0
Prediction=1
Suppose prediction was 0.
Error:

$$1 - 0 = 1$$

Update Weight

$$W_1 = 0.5 + 0.1(1)(2)$$

=0.7

$$W_2 = 0.2 + 0.1(1)(3)$$

=0.5
Bias:

$$0.1 + 0.1 = 0.2$$

Updated weights:
(0.7,0.5)
Bias=0.2

Advantages

- Easy to implement
- Fast learning
- Suitable for linearly separable data

Limitations

- Cannot solve XOR problem
- Works only for binary classification

Applications

- Spam Detection
- Face Detection
- Pattern Recognition

Conclusion

The Perceptron learns by adjusting weights and bias whenever prediction errors occur. It forms the foundation of modern neural networks.

Q4.
Illustrate the forward propagation process in a Multilayer Perceptron (MLP) using a simple neural network with one hidden layer. **Compute** the output values for the given inputs, weights, biases, and activation function. (10 marks)

ANS:

Introduction

Forward propagation is the process of passing input data through the neural network to compute the final output.

MLP Architecture

Input Layer
|
Hidden Layer
|
Output Layer

Given

Inputs:

$X_1=1$

$X_2=2$

Hidden Weights:

$W_1=0.5$

$W_2=0.4$

Hidden Bias=0.2

Activation=Sigmoid

Output Weight=0.6

Output Bias=0.1

Step 1: Hidden Layer Calculation

Weighted Sum

$$Z = (1 \times 0.5) + (2 \times 0.4) + 0.2$$

=0.5+0.8+0.2

=1.5

Apply Sigmoid

$$A = \frac{1}{1 + e^{-1.5}}$$

≈0.818

Step 2: Output Layer

$$Z_o = (0.818 \times 0.6) + 0.1$$

=0.5908

Apply Sigmoid

$$Y = \frac{1}{1 + e^{-0.5908}}$$

≈0.643

Final Output:

0.643

Working

1. Inputs are multiplied by weights.
2. Bias is added.
3. Activation function generates hidden outputs.
4. Hidden outputs are sent to output layer.
5. Final prediction is produced.

Applications

- Image Classification
- Speech Recognition
- Fraud Detection

Conclusion

Forward propagation calculates the network output layer by layer using weights, bias, and activation functions. It is the first step in neural network training.

Q5.

Demonstrate the Backpropagation algorithm for training a neural network by calculating the output error, propagating the error through hidden layers, and updating the network weights using Gradient Descent. (10 marks)

ANS:

Introduction

Backpropagation is a supervised learning algorithm used to train neural networks. It minimizes prediction error by updating the weights and biases using **Gradient Descent**.

Steps in Backpropagation

Step 1: Forward Propagation

Pass the input through the network and compute the predicted output.

Suppose:

Target = 1

Predicted Output = 0.7

Step 2: Calculate Output Error

$$\begin{aligned} &Error = Target - Prediction \\ &= 1 - 0.7 \\ &= 0.3 \end{aligned}$$

Step 3: Compute Gradient

The gradient indicates the direction in which the weights should be updated to reduce the error.

Step 4: Update Weights Using Gradient Descent

Weight update formula:

$$W_{new} = W + \eta(Error)$$

Given:

Weight = 0.5

Learning Rate = 0.1

$$\begin{aligned} &W_{new} = 0.5 + (0.1 \times 0.3) \\ &= 0.53 \end{aligned}$$

Bias update:

$$b_{new} = 0.2 + (0.1 \times 0.3) = 0.23$$

Step 5: Propagate Error to Hidden Layer

The output error is propagated backward to the hidden layer. Hidden layer weights are also updated using the calculated gradients.

Step 6: Repeat Until Convergence

The process of forward propagation, error calculation, backpropagation, and weight updates continues for many epochs until the error becomes minimal.

Advantages

- Improves prediction accuracy
- Efficiently trains deep neural networks
- Reduces loss function through optimization

Applications

- Image Recognition
- Speech Recognition
- Medical Diagnosis
- Natural Language Processing (NLP)
- Autonomous Vehicles

Conclusion

Backpropagation is the most widely used algorithm for training neural networks. By propagating errors backward and updating weights using Gradient Descent, it enables the network to learn from data and make accurate predictions.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach

Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _______________ Date: ___________________	Signature: _______________ Date: ___________________	Signature: _______________ Date: ___________________